M. Sc. (I.C.T.) 2nd Semester

Course: ICT 203 Elective 1: Smart Device Computing Using iOS

Course Code	ICT 203 Elective 1
Course Title	Smart Device Computing Using iOS
Credit	4
Teaching per Week	4 Hrs
Minimum weeks per Semester	15 (Including Class work, examination, preparation, holidays etc.)
Last Review / Revision	June 2023
Purpose of Course	The Purpose of course is to help understanding the components and structure of mobile application development using iOS. The course also provides students with the skills necessary to develop an iOS App from scratch to deploying it on the Apple Store.
Course Objective	The objective of the course is to impart knowledge of Swift and Apple iOS application Design and Development.
Course Outcomes	CO1 : Students will be able to understand Apple based smart device application development CO2 : Students will be able to learn about various components of iOS application development tools CO3 : Students will be able to publish iOS application on Apple store.
Mapping between COs with PSOs	(see table below)
Pre-requisite	Object Oriented Programming knowledge.

Mapping between COs with PSOs

	PSO1	PSO2	PSO3	PSO4	PSO5
CO1					
CO2					
CO3					

Course Content

Unit 1 : Introduction to iOS with Swift Language

1.1 Introduction iOS and iOS Architecture
1.1.1. Foundation Framework
1.1.2. Cocoa Framework
1.2 Introduction to Xcode IDE
1.2.1. Setting up Development Environment
1.2.2. Xcode Development Tools – Interface Builder and Simulator
1.2.3. Testing and Debugging
1.3 Introduction to Swift
1.3.1. Datatypes, Variables in Swift
1.3.2. Tuples, Constants, Literals in Swift
1.3.3. Working with Strings in Swift
1.4 Optionals in Swift – Implicit and Explicit
1.5 Collections in Swift
1.5.1. Dictionaries, Arrays, and Sets
1.6 Control Flows and Functions in Swift
1.7 Object Oriented Programming in Swift
1.7.1. Custom Class and Instance Creation
1.7.2. Inheritance and Polymorphism
1.7.3. Initializers in swift
1.8 Protocols and Extensions
1.9. Information Property List File and App Permissions

Unit 2 : iOS Design Patterns

2.1 Introduction to Storyboard
2.2 Introduction to UIView, UIWindow and UIViewController
2.3 Model View Controller (MVC) Pattern in Interface Design
2.4 Application Life Cycle and View Controller Life Cycle
2.5 Storyboard and Interface builder
2.6 Working with Basic UIElements
2.6.1 UILabel, UIButton, UITextFeild, UIImageView etc.
IBActions and IBOutlets

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Segue

Alert
Actionsheet

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2.8 Auto Layout Constraints to create Adaptive UI
2.9 UIAnimation
2.9.1 Animation using Auto Layout Constraints
2.9.2 Animation with UIImageView
2.10 Recognizing and Handling Gestures
2.10.1 Working with different types of Gestures
2.10.2 Gestures with UIElements

Unit 3 : UIControls in iOS

3.1 Navigation Controller and its Usage
3.2 Navigation Techniques
Segue, Push, Pop, Present and Dismiss
Working with TableView
Static TableViewController
3.3.2 Dynamic TableView
Working with PickerView
3.5 Working with Miscellaneous Controls in iOS
3.5.1 UICollectionView
3.5.2 UITabBarController
3.5.3 UIScrollView
3.5.4 UIWebView
3.5.5 ContainerView
Working with AlertController and its Types

Unit 4 : Data Persistence and Data Manipulation Techniques

~~Working with UserDefaults for data persistence~~
4.2 Introduction to FileManager
4.3 Frameworks and Library Configurations
Data Persistence Techniques
~~SQlite Framework~~
~~Core Data Framework~~
4.5 Data Manipulation Techniques
4.5.1 JSON Parsing
4.5.2 XML Parsing

Unit 5 : Advance Programming in iOS

5.1 API intergation
5.2 Location based Services
5.1.1. Core Location Services
5.1.2. CLLocation and CLLocationManager Classes
5.1.3. MapKit, MapView and MKPointAnnotation
5.1.4. Location Based Call-outs
5.3 Introduction to the working of Push Notifications
5.4 Publishing iOS App to Apple Store
5.5 Introduction to CoreML
5.6 Introduction to SwiftUI

Reference Book:	1. Swift Programming: The Big Nerd Ranch Guide (2nd Edition) (Big Nerd Ranch Guides) 2nd Edition by Matthew Mathias (Author), John Gallagher (Author) 2. Swift: A Comprehensive Intermediate Guide to Learn and Master the Concept of Swift Programming Kindle Edition by MG Martin (Author) 3. iOS 12 Programming Fundamentals with Swift: Swift, Xcode, and Cocoa Basics 1st Edition by Matt Neuburg (Author) 4. Classic Computer Science Problems in Swift: Essential Techniques for Practicing Programmers1st Edition by David Kopec 5. iOS Programming: The Big Nerd Ranch Guide, by Christian Keur and Aaron Hillegass 6. Beginning Swift by Rob kerr and Kare Morstol, Packt Publication
Teaching Methodology:	Lectures, Discussion, Independent Study, Hands-on-Session, Seminars and Assignment
Evaluation Method:	30% Internal assessment 70% External assessment